BIG DATA ANALYTICS FOR E-COMMERCE

Customer Behavior Analysis and Product Recommendation System

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Big Data Analytics BCSE402L

GitHub Repository link: https://github.com/Aarav3208/E-Com-BigData

Item	Details
Project status	Current review milestone completed
Primary technology	Python + PySpark
Input dataset	Online Retail II transaction CSV
Execution mode	Local Spark on Windows
Distributed storage	HDFS utility prepared
Processed format	Parquet

1. Executive Summary

This project implements a working Big Data analytics pipeline for e-commerce transaction behavior. The current milestone covers dataset ingestion, data cleaning, revenue calculation, feature engineering, customer and product aggregation, local Parquet storage, optional HDFS support, and Spark SQL analytics.

The recommendation engine, MongoDB integration, Power BI dashboard, APIs, cloud services, streaming, orchestration, and deployment are intentionally excluded from this review milestone.

2. Project Objective

The objective is to analyze customer purchasing behavior and product performance using a distributed-data-oriented architecture. The pipeline prepares reliable analytical datasets that can later support recommendation generation and dashboard reporting.

3. Architecture

```
Online Retail II CSV
        |
        v
Data loading with PySpark
        |
        v
Data cleaning and validation
        |
        v
Feature engineering and aggregates
        |
        +--> Processed Parquet datasets
        |
        +--> Optional HDFS raw/processed/analytics paths
        |
        v
Spark SQL analytics
```

```
        |
        v
CSV analytical output directories
```

- Raw dataset is loaded as a PySpark DataFrame.
- Invalid, incomplete, duplicate, cancelled, and non-positive transaction records are removed.
- Transactions are enriched with revenue and date/time features.
- Customer-level and product-level aggregate datasets are written as Parquet.
- Spark SQL creates ten analytical result sets under data/output/.

4. Implemented Project Structure

Path	Purpose
main.py	Command-line entry point for load, clean, features, hdfs, analytics, and all
src/config.py	Project paths, Spark settings, HDFS settings, and local Hadoop support paths
src/spark_session.py	Reusable SparkSession and Windows local filesystem configuration
src/data_loader.py	CSV discovery, loading, schema display, sample display, and row count
src/data_cleaner.py	Column normalization, validation, cancellation handling, and revenue
src/feature_engineering.py	Transaction features and customer/product aggregates
src/hdfs_manager.py	Optional HDFS directory creation, upload, and verification
src/analytics.py	Ten Spark SQL queries and persisted analytical outputs
tests/	Focused tests for cleaning, revenue, and customer aggregation
README.md	Installation, usage, Windows setup, completed work, and future work
requirements.txt	PySpark and Pytest dependencies for the milestone

5. Data Loading

The loader supports CSV input as the primary format. It searches data/raw/*.csv and also supports the current root-level online_retail_II.csv as a fallback. Loading prints the source path, row count, schema, and a five-row sample.

Observed source columns:

- Invoice
- StockCode
- Description
- Quantity
- InvoiceDate
- Price
- Customer ID

- Country

6. Data Cleaning

- Duplicate records removed.
- Rows without Customer ID removed.
- Rows without StockCode removed.
- Non-positive Quantity removed.
- Non-positive Price removed.
- Cancelled invoices identified by invoice values beginning with C and removed.
- InvoiceDate converted to a Spark timestamp.
- Column Customer ID normalized to CustomerID.
- Revenue calculated as Quantity multiplied by Price.
- Before and after row counts printed for demonstration.

Cleaning stage	Observed count
Input records	1,067,371
Cleaned records	779,425

7. Feature Engineering and Aggregation

Transaction-level fields created:

- Revenue
- Year
- Month
- Day
- Hour
- DayOfWeek

Customer-level fields:

- CustomerID
- TotalOrders
- TotalQuantity
- TotalRevenue
- UniqueProducts
- AverageOrderValue

Product-level fields:

- StockCode
- Description
- TotalQuantitySold
- TotalRevenue
- UniqueCustomers

8. Storage Outputs

The pipeline writes the following processed datasets as Spark Parquet directories:

- data/processed/transactions
- data/processed/customers
- data/processed/products

Spark writes part files and a _SUCCESS marker inside each directory. These are directory-based Parquet datasets and should be read with Spark using the directory path.

9. Spark SQL Analytics

1. Total revenue
2. Total orders
3. Total customers
4. Total products
5. Revenue by country
6. Top 10 products by revenue
7. Top 10 products by quantity sold
8. Monthly revenue
9. Top customers by revenue
10. Average order value

Each result is displayed in the terminal and written under data/output/ as a Spark CSV output directory.

10. Windows Spark Runtime Fix

The original Windows failure occurred when Spark attempted to write Parquet. Hadoop's RawLocalFileSystem tried to invoke winutils.exe and raised a FileNotFoundException because HADOOP_HOME and hadoop.home.dir were unset and no winutils.exe was available.

The installed environment was inspected before the fix:

Component	Detected version/details
PySpark	3.5.9
Bundled Hadoop client API	3.3.4
Bundled Hadoop client runtime	3.3.4
Java	IntelliJ IDEA 2025.1.3 JBR

A random or mismatched winutils.exe was not installed. Instead, the project uses the verified pure-Java GlobalMentor local filesystem adapter version 0.1.0. It is stored in the ignored project-local path hadoop-local/lib and configured automatically by src/spark_session.py on Windows.

- HADOOP_HOME is set to the project-local hadoop-local directory for the process.
- hadoop.home.dir is configured through Spark.
- The BareLocalFileSystem class is selected for file:// operations.
- The adapter JAR is added to Spark driver and executor classpaths.
- No HDFS cluster is installed or required for local Spark and Parquet.

- The binary support directory is ignored by Git.

11. Commands

PowerShell setup:

```
$env:JAVA_HOME="C:\Program Files\JetBrains\IntelliJ IDEA 2025.1.3\jbr"
$env:Path="$env:JAVA_HOME\bin;$env:Path"
.\.venv\Scripts\Activate.ps1
```

Install dependencies:

```
python -m pip install -r requirements.txt
```

Pipeline commands:

```
python main.py load
python main.py clean
python main.py features
python main.py hdfs
python main.py analytics
python main.py all
```

Validation commands:

```
python -m compileall -q main.py src tests
python -m pytest -q
```

12. Validation Results

Check	Result
Python compilation	Passed
PySpark tests	3 passed
Full main.py all run	Completed successfully
Feature engineering Parquet writes	Completed
Spark SQL analytics	Completed
HDFS CLI	Optional and unavailable; handled non-fatally

The full run passed the previous Windows failure point after feature engineering, successfully wrote transactions, customers, and products, and continued through all Spark SQL analytics.

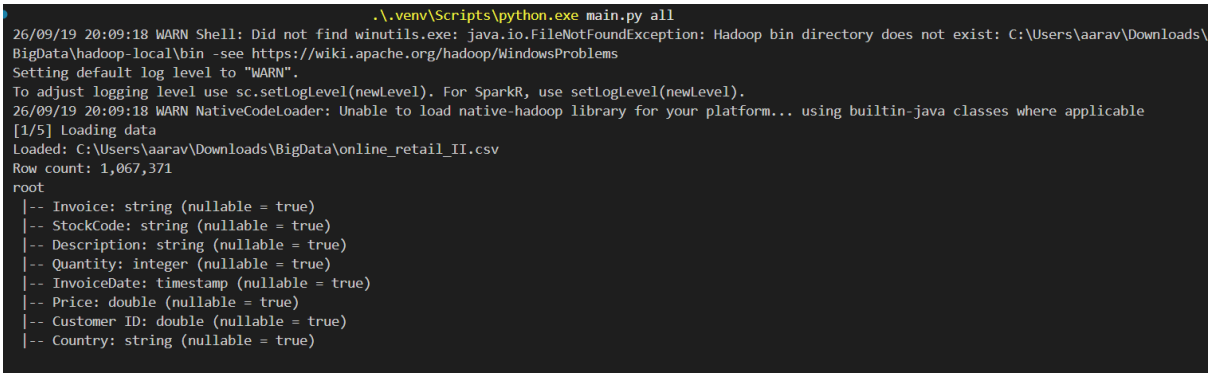
13. Future Work

11. Collaborative filtering recommendation engine using Spark MLlib ALS.
12. Recommendation evaluation metrics.
13. MongoDB recommendation storage.
14. Customer segmentation and RFM analysis.
15. Power BI dashboard.
16. Final end-to-end deployment.

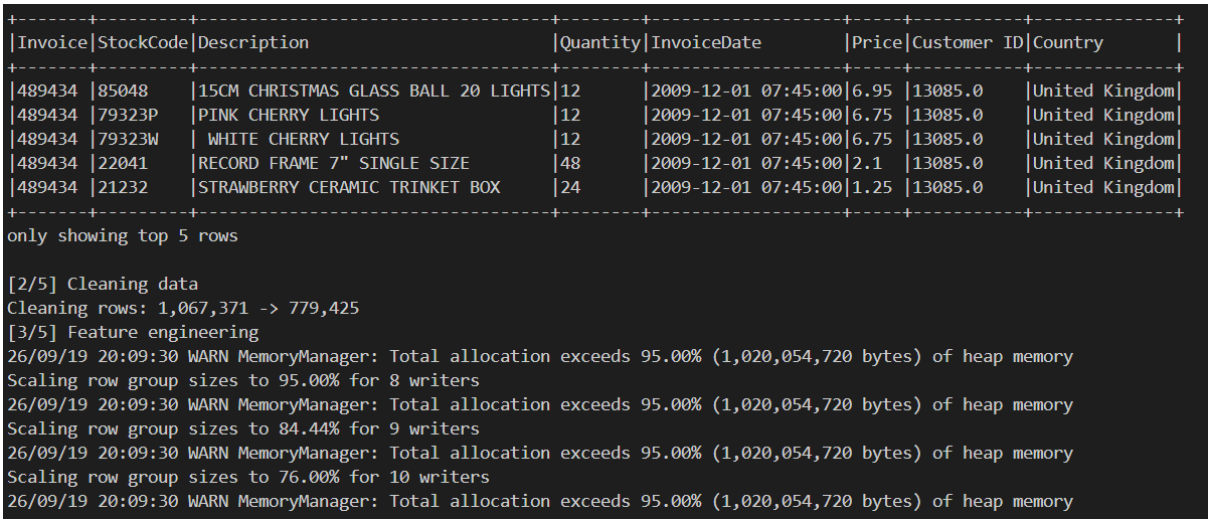
14. Current Scope Boundary

The following were intentionally not implemented in this milestone: recommendation engine, MongoDB integration, Power BI/dashboard, FastAPI, React, AWS, Kafka, Airflow, Kubernetes, and a full Hadoop/HDFS cluster installation.

15. Output Screenshots:



```
.\.venv\Scripts\python.exe main.py all
26/09/19 20:09:18 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: Hadoop bin directory does not exist: C:\Users\aarav\Downloads\
BigData\hadoop-local\bin -see https://wiki.apache.org/hadoop/WindowsProblems
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/09/19 20:09:18 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
[1/5] Loading data
Loaded: C:\Users\aarav\Downloads\BigData\online_retail_II.csv
Row count: 1,067,371
root
 |-- Invoice: string (nullable = true)
 |-- StockCode: string (nullable = true)
 |-- Description: string (nullable = true)
 |-- Quantity: integer (nullable = true)
 |-- InvoiceDate: timestamp (nullable = true)
 |-- Price: double (nullable = true)
 |-- Customer ID: double (nullable = true)
 |-- Country: string (nullable = true)
```



```
+-------+---------+-----------------------------------+--------+-------------------+-----+-----------+--------------+
|Invoice|StockCode|Description                        |Quantity|InvoiceDate        |Price|Customer ID|Country       |
+-------+---------+-----------------------------------+--------+-------------------+-----+-----------+--------------+
|489434 |85048    |15CM CHRISTMAS GLASS BALL 20 LIGHTS|12      |2009-12-01 07:45:00|6.95 |13085.0    |United Kingdom|
|489434 |79323P   |PINK CHERRY LIGHTS                 |12      |2009-12-01 07:45:00|6.75 |13085.0    |United Kingdom|
|489434 |79323W   | WHITE CHERRY LIGHTS               |12      |2009-12-01 07:45:00|6.75 |13085.0    |United Kingdom|
|489434 |22041    |RECORD FRAME 7" SINGLE SIZE        |48      |2009-12-01 07:45:00|2.1  |13085.0    |United Kingdom|
|489434 |21232    |STRAWBERRY CERAMIC TRINKET BOX     |24      |2009-12-01 07:45:00|1.25 |13085.0    |United Kingdom|
+-------+---------+-----------------------------------+--------+-------------------+-----+-----------+--------------+
only showing top 5 rows

[2/5] Cleaning data
Cleaning rows: 1,067,371 -> 779,425
[3/5] Feature engineering
26/09/19 20:09:30 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 95.00% for 8 writers
26/09/19 20:09:30 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 84.44% for 9 writers
26/09/19 20:09:30 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
Scaling row group sizes to 76.00% for 10 writers
26/09/19 20:09:30 WARN MemoryManager: Total allocation exceeds 95.00% (1,020,054,720 bytes) of heap memory
```

```
[4/5] HDFS storage
HDFS command not found. Start Hadoop and add its bin directory to PATH.
HDFS command not found. Start Hadoop and add its bin directory to PATH.
HDFS command not found. Start Hadoop and add its bin directory to PATH.
HDFS command not found. Start Hadoop and add its bin directory to PATH.
HDFS verification:
HDFS command not found. Start Hadoop and add its bin directory to PATH.
[5/5] Spark SQL analytics

total_revenue
+-------------+
|TotalRevenue |
+-------------+
|1.737480427E7|
+-------------+


total_orders
+-----------+
|TotalOrders|
+-----------+
|36969      |
+-----------+
```

```
+--------------+
|TotalCustomers|
+--------------+
|5878          |
+--------------+


total_products
+-------------+
|TotalProducts|
+-------------+
|4631         |
+-------------+
```

revenue_by_country

Country	TotalRevenue
United Kingdom	1.438923492E7
EIRE	616570.54
Netherlands	554038.09
Germany	425019.71
France	348768.96
Australia	169283.46
Spain	108332.49
Switzerland	100061.94
Sweden	91515.82
Denmark	68580.69

only showing top 10 rows

top_products_by_revenue

StockCode	Description	TotalRevenue
22423	REGENCY CAKESTAND 3 TIER	277656.25
85123A	WHITE HANGING HEART T-LIGHT HOLDER	247203.36
23843	PAPER CRAFT , LITTLE BIRDIE	168469.6
85099B	JUMBO BAG RED WHITE SPOTTY	167920.64
M	Manual	151777.67
POST	POSTAGE	124648.04
84879	ASSORTED COLOUR BIRD ORNAMENT	124351.86
47566	PARTY BUNTING	103283.38
23166	MEDIUM CERAMIC TOP STORAGE JAR	81416.73
22086	PAPER CHAIN KIT 50'S CHRISTMAS	76598.18

top_products_by_quantity

StockCode	Description	TotalQuantitySold
84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS	105185
85099B	JUMBO BAG RED WHITE SPOTTY	93436
85123A	WHITE HANGING HEART T-LIGHT HOLDER	91814
21212	PACK OF 72 RETRO SPOT CAKE CASES	89850
23843	PAPER CRAFT , LITTLE BIRDIE	80995
84879	ASSORTED COLOUR BIRD ORNAMENT	78234
23166	MEDIUM CERAMIC TOP STORAGE JAR	77916
22197	SMALL POPCORN HOLDER	76944
17003	BROCADE RING PURSE	70082
21977	PACK OF 60 PINK PAISLEY CAKE CASES	54592

```
monthly_revenue
+----+-----+------------+
|Year|Month|TotalRevenue|
+----+-----+------------+
|2009|12   |683504.01   |
|2010|1    |555802.67   |
|2010|2    |504558.96   |
|2010|3    |696978.47   |
|2010|4    |591982.0    |
|2010|5    |597833.38   |
|2010|6    |636371.13   |
|2010|7    |589736.17   |
|2010|8    |602224.6    |
|2010|9    |829013.95   |
+----+-----+------------+
only showing top 10 rows
```

```
+----------+------------+
|CustomerID|TotalRevenue|
+----------+------------+
|18102.0   |580987.04   |
|14646.0   |528602.52   |
|14156.0   |313437.62   |
|14911.0   |291420.81   |
|17450.0   |244784.25   |
|13694.0   |195640.69   |
|17511.0   |172132.87   |
|16446.0   |168472.5    |
|16684.0   |147142.77   |
|12415.0   |144458.37   |
+----------+------------+
```

```
average_order_value
+-----------------+
|AverageOrderValue|
+-----------------+
|469.98           |
+-----------------+

(.venv) PS C:\Users\aarav\Downloads\BigData> SUCCESS: The process with PID 9028 (child process of PID 20668) has been terminated.
SUCCESS: The process with PID 20668 (child process of PID 31676) has been terminated.
SUCCESS: The process with PID 31676 (child process of PID 26024) has been terminated.
```